

Translated

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Binary Exponentiation

Binary exponentiation (also known as exponentiation by squaring) is a trick which allows to calculate a^n using only $O(\log n)$ multiplications (instead of $O(n)$ multiplications required by the naive approach).

It also has important applications in many tasks unrelated to arithmetic, since it can be used with any operations that have the property of **associativity**:

$$(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$$

Most obviously this applies to modular multiplication, to multiplication of matrices and to other problems which we will discuss below.

Algorithm

Raising a to the power of n is expressed naively as multiplication by a done $n - 1$ times: $a^n = a \cdot a \cdot \dots \cdot a$. However, this approach is not practical for large a or n .

$$a^{b+c} = a^b \cdot a^c \text{ and } a^{2b} = a^b \cdot a^b = (a^b)^2.$$

The idea of binary exponentiation is, that we split the work using the binary representation of the exponent.

Let's write n in base 2, for example:

$$3^{13} = 3^{1101_2} = 3^8 \cdot 3^4 \cdot 3^1$$

Since the number n has exactly $\lfloor \log_2 n \rfloor + 1$ digits in base 2, we only need to perform $O(\log n)$ multiplications, if we know the powers $a^1, a^2, a^4, a^8, \dots, a^{2^{\lfloor \log n \rfloor}}$.

So we only need to know a fast way to compute those. Luckily this is very easy, since an element in the sequence is just the square of the previous element.

$$\begin{aligned} 3^1 &= 3 \quad 3^2 = \left(3^1\right)^2 = 3^2 = 9 \quad 3^4 = \left(3^2\right)^2 = 9^2 \\ &= 81 \quad 3^8 = \left(3^4\right)^2 = 81^2 = 6561 \end{aligned}$$